

Linear Algebra

Personal Notes

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1 The Geometry of Linear Equations

The basic computational problem of linear algebra is to solve a *system of linear equations*. A system with n equations and n unknowns can be understood in three complementary ways: one equation at a time, one column at a time, or as a single matrix equation.

1.1 A two-dimensional system

Consider

$$\begin{aligned}2x - y &= 0, \\ -x + 2y &= 3.\end{aligned}$$

In the *row picture*, each equation describes a line in the xy -plane. The first line passes through the origin and the point $(1, 2)$; the second contains $(-3, 0)$ and $(-1, 1)$. Their intersection is $(1, 2)$, so the simultaneous solution is

$$x = 1, \quad y = 2.$$

Substitution checks both equations. This picture makes the meaning of a solution immediate: it is a point satisfying every row at once.

The same system has a different geometric interpretation in the *column picture*. Grouping coefficients by unknown gives

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

A *linear combination* of vectors $c_1, \dots, c_k$ is an expression $a_1c_1 + \dots + a_kc_k$ with scalar coefficients a_i . Solving the system therefore means choosing the combination of the two coefficient columns that produces the right-hand side. Here

$$1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix},$$

which again yields $x = 1$ and $y = 2$.

1.2 The matrix equation

The compact algebraic form is

$$\underbrace{\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 0 \\ 3 \end{bmatrix}}_{\mathbf{b}}, \quad A\mathbf{x} = \mathbf{b}.$$

The matrix A is the *coefficient matrix*, $\mathbf{x}$ is the vector of unknowns, and $\mathbf{b}$ is the right-hand-side vector.

The *matrix–vector product* $A\mathbf{x}$ should first be viewed as a combination of the columns of A . If

$$A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

then

$$A\mathbf{x} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}.$$

Equivalently, each component is the dot product of a row of A with $\mathbf{x}$:

$$A\mathbf{x} = \begin{bmatrix} 2 \cdot 1 + 5 \cdot 2 \\ 1 \cdot 1 + 3 \cdot 2 \end{bmatrix}.$$

The row calculation and column combination are the same multiplication seen from different directions.

1.3 Three equations and three unknowns

For the system

$$\begin{aligned} 2x - y &= 0, \\ -x + 2y - z &= -1, \\ -3y + 4z &= 4, \end{aligned}$$

the matrix form is

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}.$$

In the row picture, each equation is a plane in three-dimensional space. Two planes generally meet along a line, and a third plane can reduce that common set to one point. That point is the solution. The idea is correct but the geometry quickly becomes difficult to draw in higher dimensions.

The column picture remains algebraically useful. It asks for scalars x, y, z such that

$$x \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}.$$

Because the right-hand side is the third column, the solution is immediately $\mathbf{x} = (0, 0, 1)^T$. A different right-hand side changes the required coefficients but not the columns being combined.

1.4 Which right-hand sides are reachable?

The equation $A\mathbf{x} = \mathbf{b}$ is solvable precisely when $\mathbf{b}$ is a linear combination of the columns of A . Thus the question

“Can we solve $A\mathbf{x} = \mathbf{b}$ for every $\mathbf{b}$?”

is equivalent to asking whether the column combinations fill the entire ambient space.

An n -by- n matrix is *invertible* (or *nonsingular*) when every right-hand side has exactly one solution. Geometrically, its columns provide enough independent directions to reach any vector. By contrast, columns are *linearly dependent* when one can be formed from the others. In three dimensions, if all three columns lie in one plane, every combination remains in that plane. Right-hand sides outside the plane are unreachable, and the matrix is *singular*.

This column viewpoint scales beyond what can be drawn: even for nine equations and nine unknowns, $A\mathbf{x}$ is still a linear combination of nine columns. The next task is to develop elimination, a systematic procedure for finding the coefficients when a solution exists.

Source: MIT OpenCourseWare, 18.06SC Linear Algebra (Fall 2011), Session 1, [The Geometry of Linear Equations](#), lecture by Gilbert Strang.

2 Elimination with Matrices

Elimination solves a linear system by replacing equations with equivalent ones until the coefficient matrix has a form that is easy to solve. For an invertible matrix, the procedure produces one pivot for each unknown and then recovers the solution in reverse order.

2.1 Forward elimination

Consider

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 12 \\ 2 \end{bmatrix}, \quad A\mathbf{x} = \mathbf{b}.$$

The entry 1 in position (1, 1) is the first *pivot*. A pivot is the nonzero entry used to eliminate the entries below it. The corresponding *multiplier* for row 2 is $3/1 = 3$, so the first row operation is

$$R_2 \leftarrow R_2 - 3R_1.$$

Row 3 already has a zero beneath the first pivot. The matrix becomes

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}.$$

The second pivot is 2. Its multiplier for row 3 is $4/2 = 2$, giving

$$R_3 \leftarrow R_3 - 2R_2$$

and hence

$$A \longrightarrow U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}.$$

The matrix U is *upper triangular*: every entry below its main diagonal is zero. The pivots are 1, 2, 5.

Every row operation must also be applied to the right-hand side. Appending $\mathbf{b}$ to A creates the *augmented matrix*

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ 0 & 4 & 1 & 2 \end{array} \right].$$

The same two operations transform it into

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 5 & -10 \end{array} \right],$$

so $A\mathbf{x} = \mathbf{b}$ has become $U\mathbf{x} = \mathbf{c}$, with $\mathbf{c} = (2, 6, -10)^T$.

2.2 Back substitution and pivot failure

The triangular system is

$$\begin{aligned} x + 2y + z &= 2, \\ 2y - 2z &= 6, \\ 5z &= -10. \end{aligned}$$

Back substitution starts with the last equation and moves upward:

$$z = -2, \quad y = 1, \quad x = 2.$$

Thus $\mathbf{x} = (2, 1, -2)^T$, which also solves the original system.

The complete reduction from A to U is called *forward elimination*. A zero cannot serve as a pivot. If the pivot position contains zero but a lower entry in the same column is nonzero, a row exchange moves that entry into the pivot position. If the pivot and every entry below it are zero, no exchange can repair the step. The square matrix is then singular: some right-hand sides have no solution, while others may have infinitely many, so a unique solution for every right-hand side is impossible.

2.3 Elimination as matrix multiplication

Row operations can themselves be represented by matrices. The *elimination matrix*

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

performs $R_2 \leftarrow R_2 - 3R_1$ when it multiplies on the left:

$$E_{21}A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}.$$

The unchanged rows of E_{21} are rows of the *identity matrix*

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

which satisfies $IA = A$.

The next operation is encoded by

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}.$$

Consequently,

$$E_{32}(E_{21}A) = U.$$

Matrix multiplication is *associative*, so parentheses may move without changing the order of the factors:

$$E_{32}(E_{21}A) = (E_{32}E_{21})A.$$

The product $E = E_{32}E_{21}$ performs both elimination steps. Applying the same matrix to both sides gives

$$E(A\mathbf{x}) = E\mathbf{b} \iff (EA)\mathbf{x} = E\mathbf{b} \iff U\mathbf{x} = \mathbf{c}.$$

The side of multiplication determines what is changed. Left multiplication combines rows, while right multiplication combines columns. Matrix multiplication is generally not commutative: PA and AP need not agree.

2.4 Row exchanges and inverses

A *permutation matrix* is obtained by rearranging rows of I . For example,

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

exchanges the first two rows of A in PA . Multiplying as AP instead exchanges the first two columns.

An *inverse* of a matrix E is a matrix E^{-1} satisfying $E^{-1}E = I$. Since E_{21} subtracts three times row 1 from row 2, its inverse adds that quantity back:

$$E_{21}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_{21}^{-1}E_{21} = I.$$

Each successful elimination step is reversible. This is why the transformed system has the same solution as the original one, and it also explains how a sequence of inverse elimination matrices can reconstruct A from U .

Source: MIT OpenCourseWare, 18.06SC Linear Algebra (Fall 2011), Session 2, [Elimination with Matrices](#), lecture by Gilbert Strang.